STATISTICAL DETECTION AND ATTRIBUTION OF AI-GENERATED AND HUMAN-WRITTEN PHISHING EMAILS USING NAIVE BAYES CLASSIFIERS

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ABSTRACT

Phishing emails increasingly incorporate content generated by large language models, which blurs the linguistic boundary between traditional human-written phishing and legitimate correspondence. This study proposes a two-stage framework for distinguishing Legitimate, Human-Written Phishing, and AI-Generated Phishing emails using statistically validated linguistic and stylometric features rather than black-box deep learning. Fourteen quantifiable features spanning lexical, syntactic, and communication-style categories – including token predictability, sentence-length variability, syntactic complexity, pronoun and urgency-keyword density, and lexical diversity – are extracted from each email. Each feature is subjected to a rigorous, non-parametric statistical pipeline: Kruskal-Wallis H-tests and Chi-Square tests screen for genuine cross-class differences, Benjamini-Hochberg correction controls the false discovery rate across simultaneous tests, and effect-size measures (Cohen's d, Cramér's V) confirm practical significance before a feature is retained. Only statistically and practically validated features proceed to classification. Two specialized Naive Bayes sub-models – a rank-binned model for continuous features and a categorical model for discrete features – independently estimate class probabilities, which are then combined through a Meta-Naive Bayes layer into a single posterior distribution. Classification proceeds in two steps: first separating legitimate from phishing emails, then, for emails flagged as phishing, attributing authorship as human-written or AI-generated using a dedicated attribution-specific feature subset validated via Mann-Whitney U and Chi-Square tests. By grounding every retained feature in formal hypothesis testing, the proposed approach offers an interpretable, computationally lightweight alternative to transformer-based detectors, while remaining explicitly traceable back to measurable linguistic evidence behind each classification decision.

Keywords: Phishing Detection, AI-Generated Text, Statistical Hypothesis Testing, Naive Bayes Classification, Stylometric Features